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Highlights

Kinematic Branch Reconfiguration of a Plane-Symmetric Spatial 7R Mechanism for Underactuated Mobile Locomotion

- A plane-symmetric spatial 7R mechanism has two one-DOF branches.
- SVD distinguishes branch singularities from ill-conditioned states.
- Singular configurations provide deterministic branch-switching gates.
- Curriculum PPO converts fitted branch coordination into locomotion.
- Physical prototype tests are specified for subsequent validation.

Kinematic Branch Reconfiguration of a Plane-Symmetric Spatial 7R Mechanism for Underactuated Mobile Locomotion

ARTICLE INFO

Keywords:
spatial 7R mechanism
branch reconfiguration
singular value decomposition
redundant actuation
curriculum learning
underactuated locomotion

ABSTRACT

This paper presents a plane-symmetric spatial 7R closed-chain mechanism for reconfigurable underactuated locomotion. Screw-system analysis and a modified Grübler–Kutzbach criterion show that the mechanism has one global degree of freedom. A passive-Jacobian formulation combined with singular value decomposition identifies two one-degree-of-freedom motion branches: a spatial symmetric branch and a degenerated planar branch. The branches intersect at singular configurations at which the instantaneous mobility increases and the active-joint assignment can be changed. Numerical continuation verifies the branch geometry and distinguishes true singularities from ill-conditioned non-singular configurations. A four-motor prototype architecture is then represented in MuJoCo and controlled through a two-stage curriculum-learning procedure. The first stage discovers the kinematic branches on a test rig and fits their joint coordination using Fourier series; the second stage uses these fitted relations as references for ground-locomotion learning. The resulting policies produce multiple locomotion modes while remaining consistent with the mechanism-level constraints. Physical experiments are planned to quantify branch switching and locomotion performance.

1. Introduction

Spatial closed-chain mechanisms combine compact topology, load-bearing capacity, and constrained spatial motion, making them attractive for reconfigurable machines and mobile robots. Geometric symmetry can further organize their configuration spaces and reduce the dimensionality of the motion that must be generated or controlled. Classical overconstrained linkages, including Bennett- and Bricard-type mechanisms, illustrate how carefully selected link dimensions and joint-axis relations can produce mobility that is not apparent from generic constraint counting. **[Citation needed]**

Singular configurations are commonly treated as undesirable because they can cause rank loss, poor transmission, and large actuator effort. For a multi-branch mechanism, however, the same configurations may also connect otherwise distinct components of the configuration space. This dual role suggests a design strategy in which a singularity is not traversed accidentally but used as a predictable gateway for selecting a motion branch. The main challenge is to establish that the branches are genuine one-degree-of-freedom motions, identify their intersections, and show that a physical actuation architecture can reliably excite them. **[Citation needed]**

Existing studies of spatial and overconstrained linkages have extensively addressed mobility, bifurcation, and singularity. In contrast, the connection between mechanism-level branch geometry and learned locomotion remains less developed. In particular, direct reinforcement learning under ground contact mixes kinematic coordination, contact dynamics, and propulsion in one optimization problem. Such coupling makes it difficult to determine whether a learned gait reflects the predicted mechanism branch or merely a contact-dependent local strategy. **[Citation needed: prior work on reconfigurable linkages and learned locomotion]**

Here, we propose a plane-symmetric spatial 7R closed-chain mechanism and use its singular configurations as deterministic gateways between a spatial symmetric branch and a degenerated planar branch. We first establish its global mobility, formulate its passive Jacobian, and derive an SVD-based criterion for branch singularities. Numerical continuation then characterizes the two branches and their intersections. Finally, a redundantly actuated prototype is evaluated in a MuJoCo-based curriculum: branch coordination is first learned on a test rig and fitted using Fourier series, after which the fitted relations guide locomotion learning on the ground. The present manuscript reports the theoretical and simulation framework; quantitative physical-prototype validation is retained as a planned experimental stage.

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2. Kinematic Modeling and Reconfiguration Analysis

2.1. Mechanism architecture and geometric parameters

The proposed mechanism is a spatial single-loop chain containing seven revolute joints. Its geometry is symmetric with respect to a plane that maps the corresponding links and joint axes on the two sides of the chain. The joint variables are denoted by $\mathbf{q} = [q_1, q_2, \dots, q_7]^T$. Joint 2 is selected as the input for tracking the spatial symmetric branch, whereas Joint 1 is used to parameterize the degenerated planar branch.

The plane-symmetric geometry is defined by the structural relations

$$\begin{aligned} &\alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = \alpha_6 = \alpha_7, \\ &a_2 = a_7, \qquad a_4 = a_5 = \frac{a_1 + a_2 + a_7}{2}, \\ &d_2 = d_6. \end{aligned} \tag{1}$$

The current design uses

$$\begin{aligned} &\alpha_1 = 0^\circ, \qquad \alpha_2 = \cdots = \alpha_7 = 90^\circ, \\ &a_1 = 40 \text{ mm}, \quad a_2 = a_7 = 80 \text{ mm}, \quad a_4 = a_5 = 100 \text{ mm}, \\ &d_2 = d_6 = 100 \text{ mm}. \end{aligned} \tag{2}$$

[Author input required: verify the complete D–H table, including the parameters not listed in Eq. (2)]

2.2. Screw-system-based mobility analysis

The joint twists are expressed in a common reference frame as $\mathbf{S}_i = [\boldsymbol{\omega}_i^T, \mathbf{v}_i^T]^T$, where $\boldsymbol{\omega}_i$ is the unit direction of the ith revolute axis and $\mathbf{v}_i = -\boldsymbol{\omega}_i \times \mathbf{r}_i$:

$$\mathbf{S}_i = \begin{bmatrix} \boldsymbol{\omega}_i \\ -\boldsymbol{\omega}_i \times \mathbf{r}_i \end{bmatrix}. \tag{3}$$

Starting from $\mathbf{S}_i^{(i)} = [0, 0, 1, 0, 0, 0]^T$ in the local joint frame, the global screw is obtained through

$$\mathbf{S}_i = \mathrm{Ad}_{\mathbf{T}_{0,i-1}} \mathbf{S}_i^{(i)}, \qquad \mathrm{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ [\mathbf{p}]_\times \mathbf{R} & \mathbf{R} \end{bmatrix}, \tag{4}$$

where $\mathbf{T} = [\mathbf{R}, \mathbf{p}; \mathbf{0}^T, 1]$.

Representative screws from the current symbolic model are

$$\begin{aligned} \mathbf{S}_1 &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{S}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -l \\ 0 \end{bmatrix}, \\ \mathbf{S}_3 &= \begin{bmatrix} \sin q_1 \\ -\cos q_1 \\ 0 \\ 0 \\ 0 \\ -a - l\cos q_1 \end{bmatrix}, \quad \mathbf{S}_4 = \begin{bmatrix} \sin q_2 \cos q_1 \\ \sin q_1 \sin q_2 \\ -\cos q_2 \\ (-a\sin q_1 + b\cos q_1)\cos q_2 \\ (a\cos q_1 + b\sin q_1 + l)\cos q_2 \\ (b + l\sin q_1)\sin q_2 \end{bmatrix}. \end{aligned} \tag{5}$$

Here, a, b, and l are reduced parameters of the symbolic model. **[Author input required: map a, b, l to the final D–H dimensions]**

Collecting the twists gives the global Jacobian

$$\mathbf{J} = \begin{bmatrix} | & | & & | \\ \mathbf{S}_1 & \mathbf{S}_2 & \cdots & \mathbf{S}_7 \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{6\times 7}. \tag{6}$$

Solving the reciprocal condition $\mathbf{J}^T\mathbf{S}^r = \mathbf{0}$ under the plane-symmetric geometry yields no non-zero common reciprocal screw. The number of common constraints is therefore $\lambda = 0$, and the order of the mechanism is

$$d = 6 - \lambda = 6. \tag{7}$$

For $n = 7$ links, $g = 7$ one-degree-of-freedom revolute joints, and $\sum f_i = 7$, the modified Grübler–Kutzbach criterion gives

$$M = d(n - g - 1) + \sum_{i=1}^{g} f_i = 6(7 - 7 - 1) + 7 = 1. \tag{8}$$

Thus, the mechanism has one global degree of freedom away from singular configurations. The complete twist expressions are reserved for the supplementary derivation because their length obscures the mobility argument.

2.3. Closed-loop kinematic formulation

Let $\mathbf{T}_i(\mathbf{q}) \in \mathrm{SE}(3)$ denote the relative transformation associated with the structural and joint transformations at Joint i. Loop closure requires

$$\mathbf{T}_1\mathbf{T}_2\mathbf{T}_3\mathbf{T}_4\mathbf{T}_5\mathbf{T}_6\mathbf{T}_7 = \mathbf{I}. \tag{9}$$

Differentiating Eq. (9) gives the velocity constraint

$$\mathbf{J}\dot{\mathbf{q}} = \mathbf{0}. \tag{10}$$

For one actuated coordinate q_a, the velocity vector is partitioned into the actuated and passive coordinates, which yields

$$\mathbf{J}_p\dot{\mathbf{q}}_p = -\mathbf{J}_a\dot{q}_a. \tag{11}$$

Here, $\mathbf{J}_p \in \mathbb{R}^{6\times6}$ is obtained by removing the column associated with the selected input from $\mathbf{J}$. Its invertibility determines whether the passive coordinates are locally unique functions of the input coordinate.

For the spatial symmetric branch, Joint 2 is the input, and the matrix blocks are explicitly

$$\mathbf{J}_a = \mathbf{S}_2, \qquad \mathbf{J}_p = \begin{bmatrix}\mathbf{S}_1 & \mathbf{S}_3 & \mathbf{S}_4 & \mathbf{S}_5 & \mathbf{S}_6 & \mathbf{S}_7\end{bmatrix}. \tag{12}$$

For the degenerated planar branch with Joint 1 as input,

$$\mathbf{J}_a = \mathbf{S}_1, \qquad \mathbf{J}_p = \begin{bmatrix}\mathbf{S}_2 & \mathbf{S}_3 & \mathbf{S}_4 & \mathbf{S}_5 & \mathbf{S}_6 & \mathbf{S}_7\end{bmatrix}. \tag{13}$$

2.4. SVD-based singularity analysis

The passive Jacobian is decomposed as

$$\mathbf{J}_p = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \qquad \mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \sigma_2, \ldots, \sigma_6), \tag{14}$$

where $\sigma_1 \geq \cdots \geq \sigma_6 \geq 0$. A true kinematic singularity occurs when

$$\mathrm{rank}(\mathbf{J}_p) < 6 \quad \Longleftrightarrow \quad \sigma_{\min}(\mathbf{J}_p) = \sigma_6 = 0. \tag{15}$$

The number of zero singular values gives the dimension added to the instantaneous null space. Accordingly, a rank loss by one raises the instantaneous mobility from one to two at the branch intersection.

In numerical calculations, a small singular value alone is insufficient to establish a bifurcation. A configuration is classified as ill-conditioned or pseudo-singular when $\sigma_{\min}$ has a local minimum but remains strictly non-zero. The condition number

$$\kappa(\mathbf{J}_p) = \frac{\sigma_1}{\sigma_6} \tag{16}$$

is used as a complementary measure of local sensitivity and transmission deterioration. This distinction prevents the ill-conditioned configurations near 90° and 270° from being misidentified as branch intersections.

2.5. Branch reconfiguration at singular configurations

Along the spatial symmetric branch, the joint variables satisfy

$$q_7 = q_1, \qquad q_6 = -q_2, \qquad q_5 = -q_3. \tag{17}$$

At a true singular configuration, the joint axes separate into the mutually orthogonal groups $\{2, 4, 6\}$ and $\{1, 3, 5, 7\}$. Holding Joints $\{2, 4, 6\}$ fixed leaves Joints $\{1, 3, 5, 7\}$ to form an equivalent planar four-bar chain. The mechanism can therefore depart from the singularity along either the spatial symmetric branch or the degenerated planar branch, depending on which joint group is actuated.

3. Numerical Validation of the Two Motion Branches

3.1. Numerical continuation and singularity identification

For a prescribed input angle, loop closure is solved by minimizing a six-dimensional residual containing the position and orientation errors of the terminal frame. A BFGS quasi-Newton solver searches for the passive joint coordinates until the residual norm is below 10^{-4}. The converged solution at the preceding input step initializes the next solve, thereby providing numerical continuation on one assembly branch and suppressing unintended jumps between branches.

With Joint 2 as the input, the singular-value spectrum is evaluated over a complete revolution. The available results identify true rank-deficient configurations near $q_2 = 0°$ and $q_2 = 180°$; $q_2 = 360°$ is physically identical to $q_2 = 0°$. Local minima near 90° and 270° remain non-zero and are therefore classified as ill-conditioned configurations rather than additional singularities.

[Evidence needed: regenerate the singularity table from the final sampling resolution and report residuals, $\sigma_{\min}$, and condition numbers consistently]

3.2. Spatial symmetric branch

The continued solutions form a smooth one-dimensional trajectory parameterized by q_2. Pairwise comparisons of the joint coordinates confirm the symmetry relations in Eq. (17). Fourier series provide a compact representation of the numerical joint trajectories,

$$\hat{q}_i(q_2) = c_{i,0} + \sum_{k=1}^{K} \left[c_{i,k} \cos(kq_2) + s_{i,k} \sin(kq_2) \right], \tag{18}$$

where the coefficients are fitted to the continued solutions. These fits are used as numerical surrogates and later as coordination references for learning; they are not treated as exact analytical closure solutions.

[Evidence needed: final Fourier order, coefficients, RMSE, maximum error, and coefficient units]

3.3. Degenerated planar branch and branch intersections

Starting from a singular configuration and selecting Joint 1 as the input tracks the degenerated planar branch. Joints 2, 4, and 6 remain fixed within numerical tolerance, whereas Joints 1, 3, 5, and 7 execute the equivalent planar four-bar motion. The existing calculations show one closed one-degree-of-freedom branch, but the complete signed angle relations for this branch must be rechecked before they are stated as exact constraints. **[Author input required: resolve the conflicting planar-branch angle relations in the source notes]**

Embedding the spatial and planar solutions in a common joint-coordinate subspace produces two one-dimensional manifolds. Their intersections coincide with the singular configurations obtained from the passive-Jacobian analysis. This agreement connects the local rank-loss criterion with the global configuration-space topology and supports deterministic branch selection at the shared configurations.

4. Prototype Design and Curriculum-Learning-Based PPO Validation

4.1. Prototype and simulation framework

The mechanism is embodied as a four-motor mobile prototype. The chassis and major links are manufactured by three-dimensional printing using polylactic-acid/carbon-fibre (PLA–CF) composite, while the left and right ground-contact links provide the primary propulsive interactions. The motors are arranged in symmetric pairs so that different

joint groups can excite the spatial branch, the planar branch, or a transition strategy. Redundant actuation is used because a single compact motor does not provide sufficient torque margin near poorly conditioned configurations.

The MuJoCo virtual prototype reproduces the CAD geometry, joint locations, motion limits, masses, inertias, and contact bodies of the physical design. This correspondence is necessary because the PPO policies are intended for the manufactured mechanism rather than for an abstract kinematic chain. **[Author input required: motor models, link masses and inertias, joint limits, friction coefficients, actuator limits, control frequency, and model identification procedure]**

PPO uses separate policy and value networks with multilayer perceptrons. The action vector commands the four motors, while the observation vector contains the joint states and the task-dependent body or contact states. Training uses 2048 parallel simulation environments and a maximum of 800 iterations in the current setup. **[Author input required: complete observation/action definitions, network widths, activation functions, PPO hyperparameters, episode duration, randomization ranges, and reward weights]**

4.2. Stage I: discovery of kinematic motion branches on the test rig

In the first curriculum stage, the virtual prototype is fixed to a test rig. Removing gross body motion and ground contact isolates the coordination problem of the closed-chain mechanism. Under a common reward definition, random seed, and PPO configuration, the six possible two-motor combinations are compared as a redundant-actuation ablation. They are grouped into plane-symmetric, spatial-symmetric, and asymmetric mixed actuation according to the selected joints.

The reward promotes the prescribed input motion, loop-compatible joint coordination, low actuation effort, and smooth actions. The ablation curves are used to compare convergence and the final motion branch rather than to claim that one policy is universally superior. Available training runs contain policies corresponding to the spatial symmetric branch, the degenerated planar branch, and branch-changing behavior near a singular configuration.

Joint trajectories from stable policies are compared with the numerical branches in Section 3. The trajectories are then fitted using Eq. (18). This step converts the discovered branch coordination into a compact reference function for the next curriculum stage. **[Evidence needed: policy-selection criterion, repeated-seed statistics, trajectory errors, correlation coefficients, and fitted coefficients]**

4.3. Stage II: Fourier-guided locomotion learning on the ground

In the second stage, the test rig is removed and the robot is placed on the ground. The main input joint follows a prescribed periodic command, and the remaining joint coordinates are encouraged to track the Fourier reference obtained in Stage I. A general reward form is

$$r = w_v r_{\text{velocity}} + w_k r_{\text{kinematic}} + w_s r_{\text{stability}} - w_e r_{\text{energy}} - w_a r_{\text{action}}, \tag{19}$$

where $r_{\text{kinematic}}$ penalizes deviations from the branch-specific Fourier coordination. The other terms promote propulsion and stable, energy-aware, smooth behavior under contact.

This curriculum separates two questions that are entangled in direct end-to-end training: whether the actuators can realize a valid kinematic branch, and whether that branch can generate useful ground motion. The current simulations produced spatial full-rotation crawling, spatial peristaltic crawling, and planar peristaltic crawling policies, including forward and reverse motion. Their quantitative comparison remains dependent on a unified evaluation protocol.

[Evidence needed: speed, displacement per cycle, step length, energy or cost of transport, tracking error, success rate, and repeated-seed statistics for each gait]

5. Physical Prototype Validation

The physical validation will test whether the branch geometry and learned coordination survive manufacturing tolerances, joint clearance, actuator dynamics, friction, impact, and unmodelled compliance. The experimental platform will record synchronized motor commands, joint angles, body pose, and electrical power while an external camera or motion-capture system measures locomotion. **[Planned experiment: specify sensors, sampling frequencies, ground surface, payload, trial duration, number of repetitions, and uncertainty estimation]**

First, the prototype will be mounted on the test rig and commanded along the spatial and planar policies. The measured joint trajectories will be compared with the numerical and MuJoCo trajectories using RMSE and maximum error. Branch switching will then be initiated near each predicted singular configuration by changing the active motor

group. Successful switching will require both departure along the target branch and a measured $\sigma_{\min}(\mathbf{J}_p)$ consistent with the singularity criterion. **[Planned experiment: test-rig branch and switching validation]**

Second, the prototype will execute each learned gait on the ground. Displacement per cycle, mean speed, step length, electrical energy, tracking error, and success rate will be reported over repeated trials. The comparison between theory, MuJoCo, and experiment will separate kinematic disagreement from contact- and actuator-related performance loss. **[Planned experiment: locomotion performance and simulation–experiment comparison]**

6. Discussion

The principal mechanism-level result is that singularity and reconfiguration are two aspects of the same configuration-space structure. Rank loss in the passive Jacobian marks configurations at which the normal one-dimensional motion ceases to be locally unique, while the orthogonal joint-axis groups provide two admissible directions for leaving the intersection. The singularity is therefore not merely a failure state; under controlled actuation it becomes a branch-selection gateway.

The learning results also indicate a useful division of labour between mechanical structure and control. The closed-chain topology restricts the admissible coordination before learning begins. Stage I identifies which redundant motor groups can excite each branch, and the Fourier fit transfers that coordination to Stage II as a reference. PPO is then used primarily to resolve contact, propulsion, energy, and stability within the mechanically admissible motion, rather than to rediscover the entire closed-chain kinematics under contact.

Several boundaries remain. The planar-branch signed angle relations require a final notation check, and the current simulation results need repeated-seed statistics and uniform gait metrics. More importantly, an accurately parameterized virtual prototype does not by itself establish transfer to the manufactured system. Joint clearance, compliance, actuator saturation, surface friction, and impact may change the attainable switching region and locomotion efficiency. For this reason, the physical experiments in Section 5 are required before making claims about real-world robustness, zero-shot generalization, steering, or terrain adaptation.

7. Conclusions

This study develops a plane-symmetric spatial 7R closed-chain mechanism whose configuration space contains a spatial symmetric branch and a degenerated planar branch. Screw-system analysis and the modified Grübler–Kutzbach criterion establish one global degree of freedom, whereas SVD of the passive Jacobian identifies the singular configurations at which instantaneous mobility increases and branch selection becomes possible. Numerical continuation verifies the two branches and separates true singularities from ill-conditioned non-singular states. A two-stage curriculum then uses test-rig PPO training to discover branch-specific coordination and Fourier-guided ground training to generate multiple locomotion modes. These results support the use of mechanism topology as a structural prior for learned locomotion, while quantitative physical prototype experiments remain necessary to establish real-world performance.

Declaration of competing interest

[Author input required: competing-interest declaration]

Data and code availability

[Author input required: repository and availability statement]
[Citation needed: verified bibliography database]

[Figure placeholder: mechanism topology, J1–J7, symmetry plane, global frame, and initial configuration]

Figure 1: Architecture and initial configuration of the plane-symmetric spatial 7R mechanism.

[Figure placeholder: SVD singularity-identification workflow and comparison of regular, ill-conditioned, and singular configurations]

Figure 2: SVD-based procedure for distinguishing true singularities from ill-conditioned non-singular configurations.

[Figure placeholder: orthogonal joint-axis groups and degeneration into the equivalent planar four-bar mechanism]

Figure 3: Kinematic interpretation of branch reconfiguration at a singular configuration.

[Figure placeholder: complete singular-value spectrum versus q_2, with true and pseudo-singular configurations annotated]

Figure 4: Singular values of the passive Jacobian along the spatial symmetric branch.

[Figure placeholder: passive joint angles versus q_2, symmetry relation plots, and Fourier-fit errors]

Figure 5: Numerical joint trajectories and symmetry relations of the spatial symmetric branch.

[Figure placeholder: planar-branch joint curves, two branch manifolds, and their intersection points]

Figure 6: Degenerated planar branch and intersections between the two one-degree-of-freedom motion manifolds.

[Figure placeholder: CAD prototype, four-motor arrangement, ground-contact links, and corresponding MuJoCo model]

Figure 7: Physical design and its MuJoCo virtual prototype.

[Figure placeholder: six-combination ablation design, reward curves, and representative test-rig configurations]

Figure 8: Redundant-actuation ablation during Stage I test-rig training.

[Figure placeholder: Fourier-guided training pipeline, learned gait sequences, and theory–simulation joint-trajectory comparisons]

Figure 9: Stage II Fourier-guided ground-locomotion learning and resulting motion modes.